LAB | Schema Design, Normalization, and Denormalization

Week 2 — Day 3: Schema Design, Normalization, and Denormalization — Lab

Part 1 — Entities, Attributes, Keys, and Relationships

Q. 1 — Find the Entities

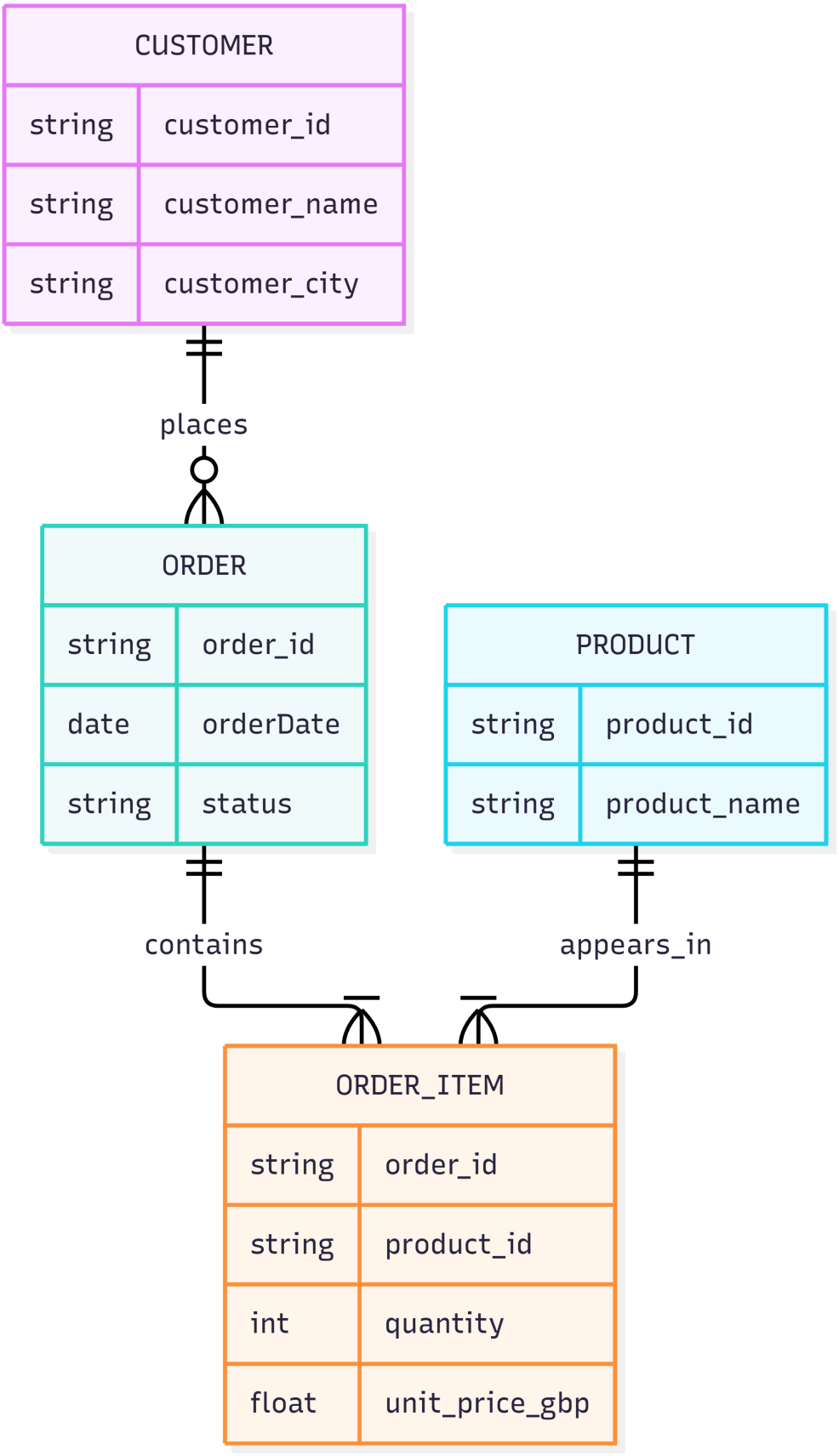
BritMart gives you this list:

> 💡 customer_id
> customer_name
> customer_city
> order_id
> order_date
> product_id
> product_name
> quantity
> unit_price_gbp

Group the attributes into these entities:

> 💡 One customer can place many orders.
> One order can contain many products.
> One product can appear in many orders.

Solution:


CUSTOMER
string
customer_id
string
customer_name
string
customer_city
places
ORDER
string
order_id
date
orderDate
string
status
PRODUCT
string
product_id
string
product_name
contains
appears_in
ORDER_ITEM
string
order_id
string
product_id
int
quantity
float
unit_price_gbp

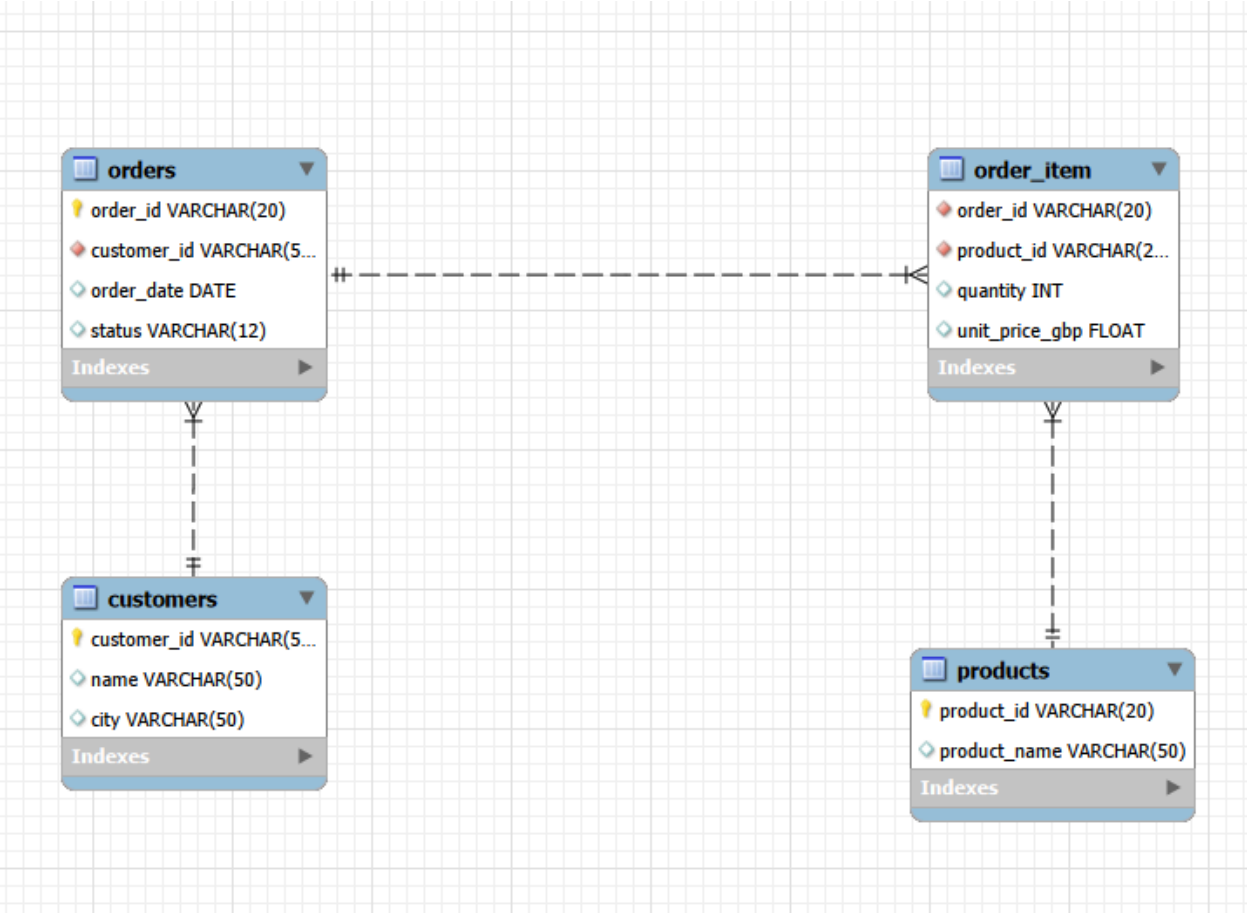
Q. 2 — Draw the Relationships

Using the four entities from Q. 1, draw these relationships:

> 💡 One customer can place many orders.
> One order can contain many products.
> One product can appear in many orders.

Use text arrows or Mermaid.

Solution:



Q. 3 — Create the Base Tables

Create MySQL tables for:

💡 customers
products
orders
order_items

Use suitable primary and foreign keys.

Solution: refer to day8_queries file.

Q. 4 — MCQ: Entity

Which item is most clearly an entity in this model?

Solution: `Product`

Part 2 — 1NF, 2NF, and 3NF

Q. 5 — Fix a 1NF Problem

This table stores more than one product in one cell:

order_id	customer_id	products
101	1	P1, P2
102	2	P3

Explain why this is a problem for 1NF.

Rewrite the data so each stored value is atomic.

Solution:

An entity is in 1NF if every attribute is a single-valued attribute. It reduces redundancy, improve data integrity and reducing anomalies in relational database design.

Below table is in 1NF.

order_id	customer_id	products
101	1	P1
101	1	P2
102	2	P3

Q. 6 — Find the 2NF Problem

Consider:

`order_id`

`product_id`

`product_name`

`quantity`

Assume the row is uniquely identified by:

`(order_id, product_id)`

`product_name` depends only on `product_id`.

Why is that a problem, and where should `product_name` it be stored?

Solution:

The problem is that product-name is partially dependent only to product_id.

2NF table:

orders
order_id
quantity
product_id

products
product_id
product_name

Q. 7 — Find the 3NF Problem

Consider:



customer_id
customer_name
postcode
city

Assume:



customer_id → postcode
postcode → city

Why can storing `city` in the same customer row create repeated data?

Suggest a 3NF-style split.

Solution:

`city` can have repeated values because it creates transitive dependancy, this increases redundancy.

3NF table:

customers→ (customer_id, customer_name, postcode)

state_postcode → (postcode, state)

Q. 8 — MCQ: Normalisation

What is the main aim of normalization in an OLTP-style model?

Solution: . B. Reduce repeated facts and keep relationships clear.

Part 3 — Normalized and Denormalized Models

Q. 9 — Insert a Small Normalized Dataset

Insert:

Customers

💡 1, Amelia Clarke, London
2, Oliver Bennett, Manchester

Products

💡 10, Desk Lamp, 25.00
11, Office Chair, 80.00
12, Notebook Set, 10.00

Orders

💡 1001, customer 1, 2026-08-01
1002, customer 2, 2026-08-01
1003, customer 1, 2026-08-02

Order Items

💡 1001, product 10, quantity 2
1001, product 12, quantity 1
1002, product 11, quantity 1
1003, product 12, quantity 3

Q. 11 — Compare Update Behavior

Imagine the product name changes from:

💡 Notebook Set

to:

💡 Premium Notebook Set

Model A

The product name is stored only once in `products`.

Model B

The product name is repeated in every order line.

Which model needs fewer data changes? Why?

Solution: Model A where the product name is stored only once. This reduces redundancy.

Q. 12 — MCQ: Denormalisation

Why might an analytical dataset deliberately repeat some descriptive values?

Solution: D. To make some analytical queries simpler to read and use

Part 4 — Scalable Model Thinking and Star Schema Introduction

A model should still make sense as data volume and the number of users grow. For analytics, a simple star schema separates measurable events from descriptive information.

Q. 13 — Identify the Fact and Dimensions

BritMart wants to analyze sales by:

> 💡 date
> customer
> product
> store

The measures are:

> 💡 quantity
> sales_gbp

Choose:

> 💡 fact table
> dimension tables

Solution:

dim_date

dim_customer

dim_product

dim_store

fact table:

dat_key

customer_id

product_id

store_id

quantity

sales_gbp

Q. 14 — Define the Grain

Choose the clearer grain for `FactSales`.

Choice A

> 💡 One row per order

Choice B

> 💡 One row per product sold within an order

BritMart needs product-level sales analysis.

Which grain should be used and why?

Solution:

Choice B is the better grain.

Q. 15 — Draw the Star

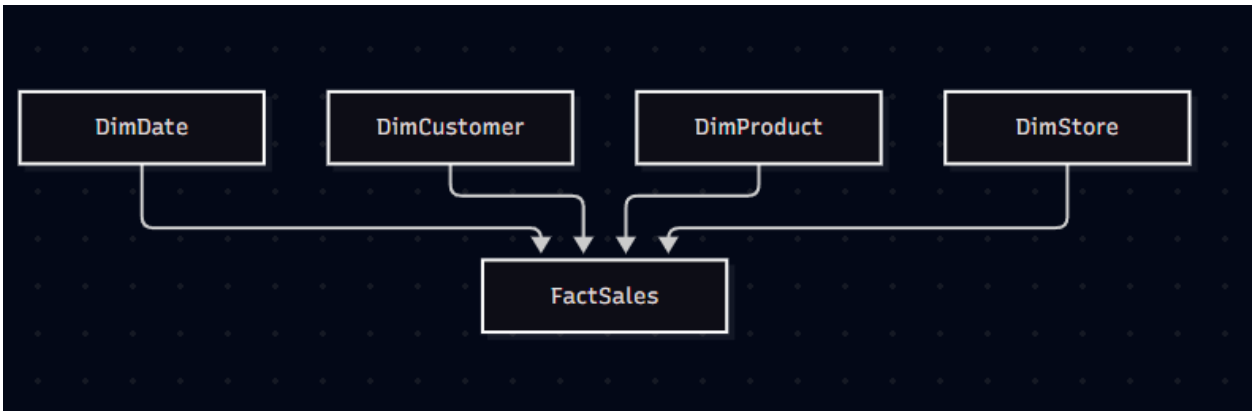
Draw:



FactSales
DimDate
DimCustomer
DimProduct
DimStore

Show the fact table in the center.

Solution:



Q. 16 — MCQ: Fact Table

Which values belong most naturally in a sales fact table?

Solution: A. `quantity` and `sales_gbp`